

## Appendix A – Troubleshooting

Use Table A-1 to help solve problems you experience with the FastBack® Model 90E. If following these steps does not correct the problem, call the Heat and Control, Inc. Service Office in Lititz, PA at (717) 519-5240 or (800) 648-5313. Have the serial number of your FastBack® available when you call. The serial number is printed on the equipment name plate (see page viii).

| Symptom                                     | Possible Cause                                                                   | Corrective Action                                                                                                                                 |
|---------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Conveyor does not run – motor does not turn | Emergency disconnect switch set to OFF position                                  | Rotate emergency disconnect switch to ON position.                                                                                                |
|                                             | Control wires not connected to correct terminals on inverter in the power module | <u>Yaskawa</u> : Connect common signal wire to terminal SC and return wire to terminal S1.<br><br>See TechPub PH009 for inverter wiring diagrams. |
|                                             | No run command from control source                                               | Troubleshoot PLC output or relay logic                                                                                                            |
| Conveyor does not run – motor turns         | Broken motor belt                                                                | Install new motor belt (see Section 5 – Maintenance)                                                                                              |
|                                             | Broken eccentric timing belt                                                     | Replace eccentric timing belt (see Section 5 – Maintenance)                                                                                       |
|                                             | Motor belt detached from motor sprocket                                          | Reinstall motor belt (see Section 5 – Maintenance)                                                                                                |
| Pan shakes excessively                      | Loose or missing band clamp                                                      | Replace band clamp                                                                                                                                |
|                                             | Cracked pan attachment disk                                                      | Replace pan attachment disk                                                                                                                       |
|                                             | Missing disk spacer                                                              | Replace disk spacer                                                                                                                               |
|                                             | Loose bolts on top of pan attachment disks                                       | Tighten bolts (see Appendix C for torque values)                                                                                                  |
|                                             | Improperly installed pan                                                         | Check pan installation (see Section 5 – Maintenance)                                                                                              |
| Enclosure shakes excessively                | Interference between slider cover and enclosure                                  | See “Slider covers scrape enclosure top” below                                                                                                    |
|                                             | Latches on enclosure loose                                                       | Tighten enclosure latches                                                                                                                         |
|                                             | Interference in slider seals (if equipped)                                       | Reinstall slider seals                                                                                                                            |
|                                             | Loose locknuts on articulating feet                                              | Tighten locknuts                                                                                                                                  |
|                                             | Threaded rods on articulating feet extend past 3 inches                          | Adjust threaded rods so that no more than 3 inches show                                                                                           |
|                                             | Interference between pan and surrounding equipment                               | Adjust equipment position to remove interference                                                                                                  |

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| Symptom                                                                             | Possible Cause                                                                        | Corrective Action                                                                  |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Slider covers scrape enclosure top                                                  | Quarter-turn latches not fully closed                                                 | Rotate latches                                                                     |
|                                                                                     | Quarter-turn latches don't completely pull enclosure down                             | Remove enclosure and adjust latch rods downward                                    |
|                                                                                     | Band clamp lever pushing down on slider cover                                         | Reorient band clamp (see "Replacing the Pan Assembly" in Section 5 – Maintenance)  |
| Conveyor runs rough                                                                 | Misadjusted timing belt                                                               | Reinstall timing belt (see Section 5 – Maintenance)                                |
|                                                                                     | Motor belt jumping                                                                    | Install new motor belt (see Section 5 – Maintenance)                               |
| Motor pulls excessively high current                                                | Imbalanced motor load – resistance between motor phases (not within 5% of each other) | Rewire motor in motor junction box                                                 |
|                                                                                     |                                                                                       | Replace motor                                                                      |
| Conveyor runs at incorrect speed                                                    | Power module not programmed for correct speed                                         | Check power module parameters and reprogram if necessary                           |
| Conveyor takes too long to start or stop                                            | Power module programmed incorrectly                                                   | Check power module parameters and reprogram if necessary                           |
| Conveyor not responding proportionately to 4-20mA signal from scale or other source | Power module programmed incorrectly                                                   | Check power module parameters and reprogram if necessary                           |
| Conveyor starting and stopping frequently (running between 1/3 and 2/3 of the time) | Infeed photoeye calibrated for too high a bed depth                                   | Reduce maximum sensing distance on photoeye by 50% and observe                     |
| Conveyor running 100% of the time, but scale is starving                            | Infeed photoeye calibrated for too low a bed depth                                    | Reduce minimum sensing distance on photoeye or check the gate feeding the conveyor |
| Conveyor running less than 1/3 of the time                                          | Infeed photoeye calibrated for too high a bed depth                                   | Reduce maximum sensing distance on photoeye by 1/4 and observe                     |
| Product spilling on floor at pan infeed                                             | Infeed photoeye calibrated for too high a bed depth                                   | Reduce maximum sensing distance on photoeye.                                       |
|                                                                                     | Pan misaligned with distribution feed point                                           | Adjust pan (see "Final Positioning" in Section 4 – Installation Procedures)        |

Table A-1 (continued) – Troubleshooting Guide

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| Symptom                                                | Possible Cause                                                   | Corrective Action                                                 |
|--------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------|
| Conveyor travel rate too low/bed depth too high        | Motor running in wrong direction                                 | Swap any two motor leads in motor junction box or at the inverter |
|                                                        | Conveyor inclined excessively                                    | Reduce incline to under 1.5°                                      |
|                                                        | Wrong preset speed selected                                      | Check and correct signal wiring or jumper connections             |
|                                                        | Incorrectly set speed reference                                  | Download initial factory settings                                 |
| Product moves the opposite direction from that desired | Incorrectly installed eccentric drive sprocket on new crankshaft | Remove key under sprocket bushing and reinstall 180° out          |
|                                                        | Drive installed backwards                                        | Replace motor<br>Turn unit around                                 |

*Table A-1 (continued) – Troubleshooting Guide*

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